1. Write a Python Program that takes two numbers as input from the user and prints addition, difference, Product and Quotient of the two numbers.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

addition = num1 + num2
difference = num1 - num2
product = num1 * num2
quotient = num1 / num2 if num2 != 0 else
"Division by zero error"

print("Addition:", addition)
print("Difference:", difference)
print("Product:", product)
print("Quotient:", quotient)
```

2. Create a class Employee with a method

calculate_salary() that returns a base salary. Define subclasses Manager and Developer that override calculate_salary() to add bonuses to the base salary. Demonstrate polymorphism by calling calculate_salary() on instances of both subclasses.

```
class Employee:
def calculate_salary(self):
return 20000

class Manager(Employee):
def calculate_salary(self):
return super().calculate_salary() + 10000
class Developer(Employee):
def calculate_salary(self):
return super().calculate_salary() + 5000

employees = [Manager(), Developer()]
for emp in employees:
print("Salary:", emp.calculate_salary())
```

1. Write a Python Program to calculate square and cube of a number.

```
num = float(input("Enter a number: "))

square = num ** 2
cube = num ** 3


print("Square:", square)
print("Cube:", cube)
```

2. Create a package named my_math with submodules arithmetic and geometry.

my_math/arithmetic.py

```
def add(a, b): return a + b
def subtract(a, b): return a - b
def multiply(a, b): return a * b
def divide(a, b): return a / b if b != 0 else "Error"
```

my_math/geometry.py

```
def rectangle_area(length, breadth): return
length * breadth
```

def rectangle_perimeter(length, breadth): return 2 * (length + breadth)

Slip 3

1. Write a Python Program to check if a number is even or odd.

```
num = int(input("Enter a number: "))
if num % 2 == 0:
print("The number is Even.")
else:
print("The number is Odd.")
```

2. Create a package named conversion with modules length and weight.

conversion/length.py

def m_to_cm(meters): return meters * 100

conversion/weight.py

def g_to_kg(grams): return grams / 1000

Slip 4

1. Write a Python Program to check if a number is Positive, Negative or Zero.

```
num = float(input("Enter a number: "))
if num > 0:
print("The number is Positive.")
elif num < 0:
print("The number is Negative.")
else:
print("The number is Zero.")
```

2. Define a base class Vehicle with a method drive() that prints "Driving a vehicle". Create derived classes Car and Motorcycle that inherit

from Vehicle. Override the drive() method in Car to print "Driving a car" and in Motorcycle to print "Riding a motorcycle".

```
class Vehicle:
def drive(self):
print("Driving a vehicle")

class Car(Vehicle):
def drive(self):
print("Driving a car")

class Motorcycle(Vehicle):
def drive(self):
print("Riding a motorcycle")

v = Vehicle()
v.drive()
c = Car()
c.drive()
m = Motorcycle()
```

```
m.drive()
```

Slip 5

1. Write a Python Program to check Leap Year.

```
year = int(input("Enter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year %
400 == 0):
print("It is a Leap Year.")
else:
print("It is not a Leap Year.")
```

2. Write a Python Program to print sum of digits of a number.

```
num = int(input("Enter a number: "))
sum_digits = 0
while num > 0:
sum_digits += num % 10
num //= 10
```

```
print("Sum of digits:", sum_digits)
```

Slip 6

1. Write a python program that ask user to enter their name and age. Print a user whose age is above 18.

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
if age > 18:
print(name, "is above 18 years of age.")
else:
print(name, "is 18 or below 18 years of age.")
```

2. Define a class Shape with a method draw() that prints "Drawing a shape". Create derived classes Rectangle and Circle that override draw() to print "Drawing a rectangle" and "Drawing a circle", respectively.

```
class Shape:
def draw(self):
print("Drawing a shape")

class Rectangle(Shape):
def draw(self):
print("Drawing a rectangle")

class Circle(Shape):
def draw(self):
print("Drawing a circle")

shapes = [Rectangle(), Circle()]
for s in shapes:
s.draw()
```

Slip 7

1. Write a python program to check entered number is divisible by 7 and multiples of 5.

```
num = int(input("Enter a number: "))
if num % 7 == 0 and num % 5 == 0:
print("Number is divisible by 7 and a multiple of 5.")
else:
print("Condition not met.")
```

2. Write a python program to print reverse of a number. (Using a function).

```
def reverse_number(n):
rev = 0
while n > 0:
rev = (rev * 10) + (n % 10)
n //= 10
return rev

num = int(input("Enter a number: "))
print("Reversed number:", reverse_number(num))
```

Slip 8

1. Write a python program to display all even numbers in between 1 to 20.

```
print("Even numbers between 1 and 20:")
for i in range(1, 21):
if i % 2 == 0:
print(i, end=" ")
```

2. Define classes A, B, and C where B and C inherit from A. Implement a method method_A() in A, and methods method_B() and method_C() in B and C respectively. Create a class D that inherits from both B and C and calls all three methods.

```
class A:
def method_A(self):
print("Method A called")
```

```
class B(A):
def method_B(self):
print("Method B called")

class C(A):
def method_C(self):
print("Method C called")

class D(B, C):
def method_D(self):
print("Method D called")

obj = D()
obj.method_A()
obj.method_B()
obj.method_C()
```

Slip 9

1. Write a python program to display those

numbers which are divisible by 3 between the range 20 and 70.

```
print("Numbers divisible by 3 between 20 and 70:")
for i in range(20, 71):
if i % 3 == 0:
print(i, end=" ")
```

2. Write a Python class Inventory with data members like item_id, item_name, stock_count and price, and methods like add_item and display_item_details.

```
class Inventory:
def init(self, item_id, item_name, stock_count, price):
self.item_id = item_id
self.item_name = item_name
self.stock_count = stock_count
self.price = price
```

```
def display_item_details(self):
print(f"ID: {self.item_id}, Name: {self.item_name},
Stock: {self.stock_count}, Price: {self.price}")

item = Inventory(101, "Pen", 50, 10.0)
item.display_item_details()
```

Slip 10

1. Write a Python Program to print Fibonacci series.

```
n = int(input("Enter number of terms: "))
a, b = 0, 1
print("Fibonacci series:")
for _ in range(n):
print(a, end=" ")
a, b = b, a + b
```

2. Create a package named shapes with modules circle and triangle. Each module should define a class with methods to calculate the area and perimeter of the respective shape.

circle.py

```
import math
class Circle:
def init(self, r): self.r = r
def area(self): return math.pi * self.r ** 2
```

triangle.py

```
class Triangle:
def init(self, a, b, c): self.a, self.b, self.c = a, b, c
def perimeter(self): return self.a + self.b + self.c
```

1. Write a Python Program to find the minimum of three numbers.

```
a = float(input("First number: "))
b = float(input("Second number: "))
c = float(input("Third number: "))

print("Minimum number:", min(a, b, c))
```

2. Create a base class Shape with a method area() that returns 0. Create derived classes Rectangle and Circle that inherit from Shape and override the area() method to calculate the area of a rectangle and circle, respectively.

```
import math

class Shape:
def area(self): return 0

class Rectangle(Shape):
```

```
def init(self, l, w): self.l, self.w = l, w
def area(self): return self.l * self.w

class Circle(Shape):
def init(self, r): self.r = r
def area(self): return math.pi * self.r ** 2

r = Rectangle(10, 5)
c = Circle(7)
print("Rectangle Area:", r.area())
print("Circle Area:", c.area())
```

Slip 12

1. Write a Python Program to print multiplication table of a given number.

```
num = int(input("Enter a number: "))
print(f"Multiplication table of {num}:")
for i in range(1, 11):
```

```
print(f"{num} x {i} = {num * i}")
```

2. Write a Python function to check whether a number is perfect or not.

```
def check_perfect(n):
sum_div = sum(i for i in range(1, n) if n % i == 0)
return sum_div == n

num = int(input("Enter a number: "))
if check_perfect(num):
print("It is a Perfect number.")
else:
print("It is not a Perfect number.")
```

Slip 13

1. Write a python program that prints all the numbers from 0 to 6 except 3 and 6. Note use continue statement.

```
for i in range(7):
if i == 3 or i == 6:
continue
print(i, end=" ")
```

2. Create a package named conversion with modules length and temperature.

length.py

def feet_to_meters(feet): return feet / 3.28084

temperature.py

def celsius_to_fahrenheit(c): return (c * 9/5) + 32

Slip 14

1. Write a Python program that calculates the

area of a circle using the math module's constant pi and function pow() for exponentiation.

```
import math
r = float(input("Enter radius: "))
area = math.pi * math.pow(r, 2)
print("Area of circle:", area)
```

2. Write a Python program to check given no is prime or not.

```
num = int(input("Enter a number: "))
if num > 1:
for i in range(2, int(num ** 0.5) + 1):
if num % i == 0:
print("Not a Prime number.")
break
else:
print("It is a Prime number.")
else:
```

```
print("Not a Prime number.")
```

Slip 15

1. Write a Python program that generates a random number between 1 and 100 using the random module.

```
import random
print("Random number:", random.randint(1, 100))
```

2. Write a Python Program to check given number is palindrome or not.

```
num = input("Enter a number/string: ")
if num == num[::-1]:
print("It is a Palindrome.")
else:
print("It is not a Palindrome.")
```

Slip 16

1. Write a python program that takes a score (0 - 100) as input from user and print corresponding grade like Distinction, First Class, Second Class, Pass class, Fail.

```
score = float(input("Enter score (0-100): "))
if score >= 75:
print("Distinction")
elif score >= 60:
print("First Class")
elif score >= 50:
print("Second Class")
elif score >= 40:
print("Pass Class")
else:
print("Fail")
```

2. Write a python program to find sum of first 'n'

natural numbers by using function.

```
def sum_natural(n):
return n * (n + 1) // 2

n = int(input("Enter n: "))
print("Sum:", sum_natural(n))
```

Slip 17

1. Write a Python Program to swap values of two variables.

```
a = input("Enter first variable: ")
b = input("Enter second variable: ")
a, b = b, a
print(f"After swapping - a: {a}, b: {b}")
```

2. Create a package named shapes with modules square and triangle. Each module should define a class with methods to calculate

the area and perimeter of the respective shape.

square.py

```
class Square:
def init(self, s): self.s = s
def area(self): return self.s ** 2
```

triangle.py

```
class Triangle:
def init(self, b, h): self.b, self.h = b, h
def area(self): return 0.5 * self.b * self.h
```

Slip 18

1. Write a python function to count all even and odd nos from a list.

```
def count_even_odd(lst):
```

```
evens = sum(1 for x in lst if x % 2 == 0)
odds = len(lst) - evens
return evens, odds

numbers = [1, 2, 3, 4, 5, 6, 7, 8]
e, o = count_even_odd(numbers)
print("Even count:", e)
print("Odd count:", o)
```

2. Write a Python Program to find factorial of a given number (Use function).

```
def factorial(n):
fact = 1
for i in range(1, n + 1):
fact *= i
return fact

num = int(input("Enter a number: "))
print("Factorial:", factorial(num))
```

Slip 19

1. Write a python program that prints only 1,2,3 the numbers from 1 to 6. Note use break statement.

```
for i in range(1, 7):
if i > 3:
break
print(i, end=" ")
```

2. Write a Python Program to print prime numbers in given interval.

```
lower = int(input("Enter lower limit: "))
upper = int(input("Enter upper limit: "))

print(f"Prime numbers between {lower} and {upper}:")
for num in range(lower, upper + 1):
```

```
if num > 1:
for i in range(2, int(num ** 0.5) + 1):
if num % i == 0:
break
else:
print(num, end=" ")
```

Slip 20

1. Write a python function to count frequency of a particular character in String.

```
def count_char(string, ch):
return string.count(ch)

s = input("Enter a string: ")
c = input("Enter character to count: ")
print("Frequency:", count_char(s, c))
```

2. Write a Python Program that takes two

numbers as input from the user and prints addition, difference, Product and Quotient of the two numbers. (Use function).

```
def perform_operations(a, b):
print("Addition:", a + b)
print("Difference:", a - b)
print("Product:", a * b)
print("Quotient:", a / b if b != 0 else "Error")

n1 = float(input("First number: "))
n2 = float(input("Second number: "))
perform_operations(n1, n2)
```